Assignment 2

Part I

Use ReducedMNIST which is a reduced version of the MNIST data set.

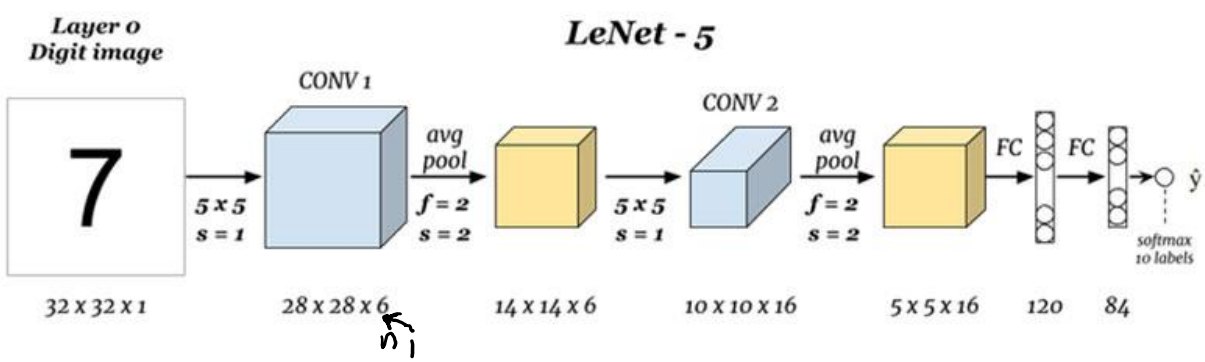
- **ReducedMNIST training**: 1000 examples for each digit to be selected randomly from the original MNIST Data.
- **ReducedMNIST test**: 200 examples for each digit.

Prob. 1:

Use multilayer perceptron (MLP) with 1, 3, or 4 hidden layers to solve this problem. You can use the DCT features used in assignment 1. You are free to use any hyper-parameters of your own choice.

Prob. 2:

a. Use the ReducedMNIST data to train CNN from the images without using a feature extraction step. Use the following structure to start with.
(Hint: you must adjust your parameters to fit 28x28 images instead of 32x32, and use **ReLU** function as an activation function).



a. Make some (at least two) variations in the hyper-parameters of your choice and check the network performance, then give your comments. These changes may be in the number of filters in any convolutional layer, the activation function, adding or removing any layer…etc.
b. Compare the results that you have obtained in this assignment with those you have obtained in Assignment 1; regarding the accuracy, the training time, and the testing time (fill the following table).

		Features						
		DCT		PCA		AutoEncoder		
		Accuracy*	Processing Time**	Accuracy	Processing Time	Accuracy	Processing Time	
Classifier								
K-means Clustering	1							DCT and PCA features from assignment 1
	4							
	16							
	32							
SVM***	Linear							
	Nonlinear:							
Multi-layer Perceptron (MLP)								Assignment 2
		DCT		PCA		AutoEncoder		
	Variations	Accuracy*	Processing Time**	Accuracy	Processing Time	Accuracy	Processing Time	
MLP	1-Hidden							
	3-Hidden							
	4-Hidden							
In the CNN Model no Features are needed								
	Variations	Accuracy	Training time		Testing time			
CNN****	Variation1:							
	Variation2:							
	Variation3:							
	Variation4:							

* Accuracy to be as % with one friction digit like 89.5% not as 89.51111111%

** Processing time in milli-seconds like 10.3 msec. not as 10.3333333 msec.

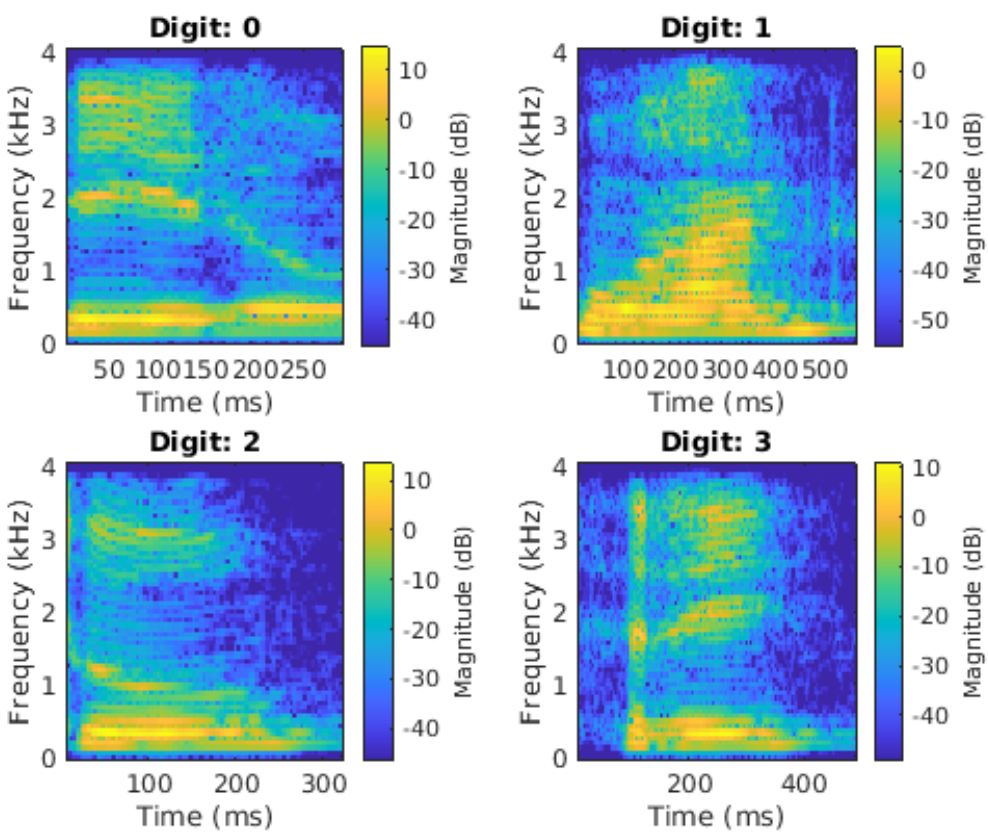
*** mention the kernel name and its specs

**** Describe each variation in short in the corresponding cells

Prob.3:

Speech Recognition from Speech Spectrum with Augmentation Experience:

a. Given the speech data (train and test) for the 10 digits uttered by many speakers, develop and train a network to recognize any given new digit using the spectrogram of each digit as the training and testing data. You may convert the speech data to images and deal with the speech problem as if it is an object recognition problem in images. **(Hint: you may start from the network in problem 1 and enhance by making some alternations in the hyper-parameters or start from any of the networks that have been used in the ImageNet problem).**
b. Repeat "a" above after making data augmentation for the speech data; for example: speed up 3% and down speed 3%, and you may add some speech noise.
c. Repeat "a" above after making data augmentation for the spectrum images; for example: squeeze (horizontally) the image by 3%, expand (horizontally) the image by 3%, and you may add some noise to the images.
d. Repeat "a" above after making speech and image augmentation as done in "b" and "c" above.



Examples of spectrum images for some digits

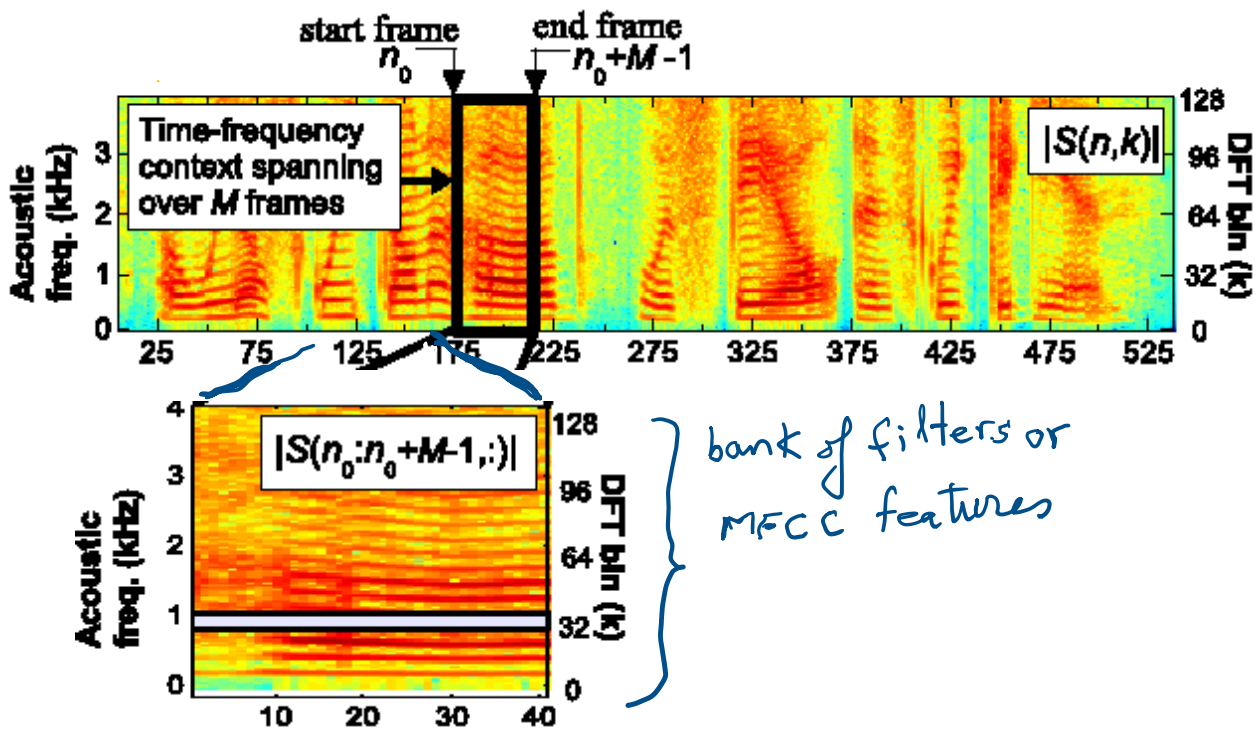
Part II

Problem 4: Data Augmentation and Data Synthesis

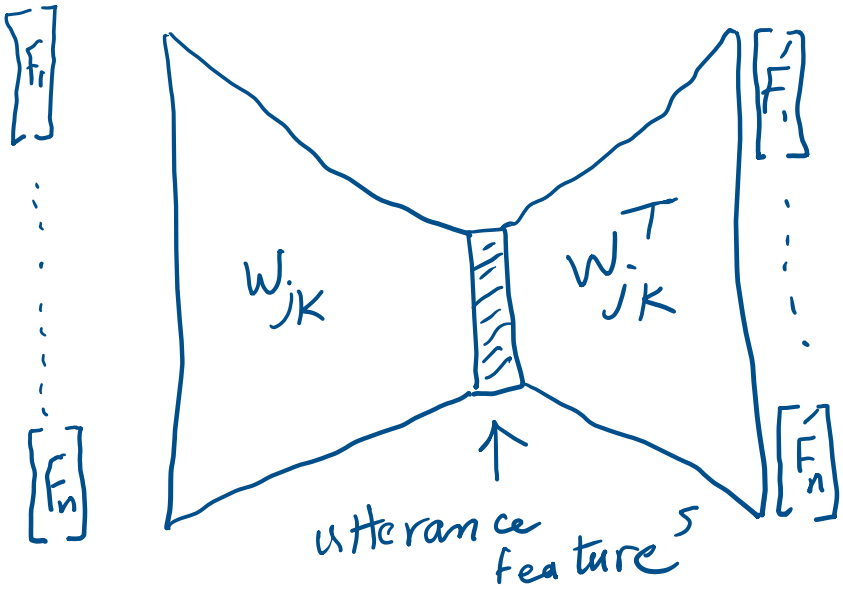
Autoencoder to represent and solve the utterances of the 10 digits given in prob 3:

Given the speech data (train and test) for the 10 digits uttered by many speakers, develop an autoencoder (AE) to generate a single vector that represents each utterance (you may divide the utterance into frames of 15 m. Sec. each, then concatenate these frames and use the AE to generate a single vector of a length of your choice to represent the complete utterance). Once you represent each utterance in a single vector, develop a classifier to classify each utterance in the test set.

1. You may start by just calculating the average frame for each utterance to get a vector that represents each utterance. Make it the baseline for your results.
2. You should try to use the AE in this way:
    a. Concatenate all the frames in one shot and generate a single vector for each utterance. However, you will have different lengths for each utterance, then use the maximum length and append zero frames in shorter utterances.



An example of a spectrum image for an utterance



Target is:

$$\min_{W_{jk}} \sum_i \| F_i - F_i' \|_2$$

An Auto Encoder is used to concatenate 2 or more frames to ultimately reduce the utterance to a single vector.

Problem 5: Data Augmentation

For the ReducedMNIST database:

1. Generate using augmentation methods different number of examples as indicated in the table below. Augmentation: include rotating right, left with different angles, moving the digit random movements by delta x and delta y, rotate each digit like 5° right or left adding % of white noise…etc.
2. Design a pipeline to test the recognition model generated with the mixture shown in the table below. (Hint: let LeNET-5 be your basic model for recognition.)
3. Fill out this table of the results for the same test data (200 examples) and add your comments on the results. (Hint: we increase the data generated as no labeling is needed gust computer time to generate.)

		Real Data for each digit		
		350	**750**	**1,000**
No. of Generated examples per digit	**0**	**350 real**	**750 real**	**1000 real**
	1,000	**350 real + 1000 generated of 350 real**	**750 real + 1000 generated of 750 real**	**1000 real + 1000 generated of 1000 real**
	1,500	**350 real + 1500 generated of 350 real**	**750 real + 1500 generated of 700 real**	**1000 real + 1500 generated of 1000 real**
	2,000	**350 real + 2000 generated of 350 real**	**750 real + 2000 generated of 750 real**	**1000 real + 2000 generated of 1000 real**

Problem 6: use GAN to generate Synthetic Data

1. Use GAN network, trained using 350 training examples only and generate 3 different examples from each digit. Comment on the generated examples regarding how close they are from the real, and the computer time exhausted in training and generation of the examples.
   (Hints:
   - use this paper to help you get a stable and reasonable starting point in your GAN implementation: https://arxiv.org/abs/1511.06434
   - It worth to visit this website too: https://towardsdatascience.com/cgan-conditional-generative-adversarial-network-how-to-gain-control-over-gan-outputs-b30620bd0cc8/
   - Add a time stamp in your code to be able to catch time information. Do not forget to mention the computer setup used.)
2. Repeat the above work but the generated data in this case will be through using the GAN network to generate each digit the number of times shown in the above table. Use a table like the one shown above and generate a model in each case. Compare the results and comment.
3. Select any reasonable combination of the augmented and synthetic data in the case of 350 examples available from real data and show to what extent we can reduce the need for real data by comparing the best case results you obtained in all the above work with the case of using all the available **1000 examples for each digit** of real data.